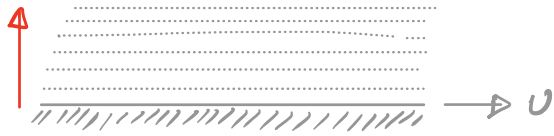


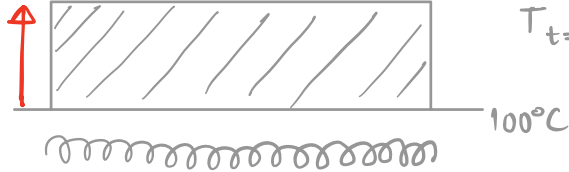
Ex:- direction of momentum transport



This transport of momentum is "molecular" and "diffusive" in nature.

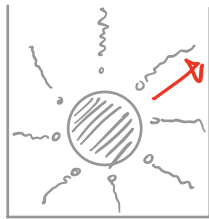
Ex:-

diffusive transport of heat energy (conduction)



$T_{t=0} = 25^\circ\text{C}$ (Initially the slab is at room temp.)

Ex:-



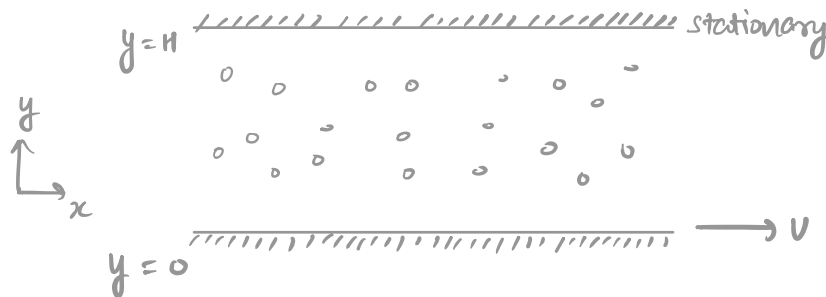
diffusive mass transfer

Therefore Transport Phenomena



(Momentum, Energy, Mass)

(Q.)



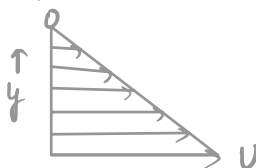
(*) Initial condition ($t=0$)

$$\forall x = 0 \quad \forall (x, y)$$

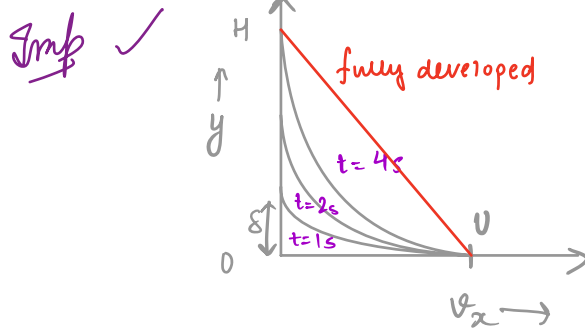
(*) Boundary Conditions

$$\left. \begin{array}{l} y=0, v_x = u \quad \forall t \\ y=H, v_x = 0 \quad \forall t \end{array} \right\} \begin{array}{l} \text{no} \\ \text{slip} \end{array}$$

→ At steady state, velocity profile will be



→ logically the developing phase can be shown as follows:-



$\delta(t)$: Momentum diffusion length scale.

δ : roughly the length scale upto which molecules are in motion or upto which momentum has transferred.

→ Cauchy - momentum balance (STUDY)

→ for newtonian fluids $\dot{\gamma}$ (shear rate)

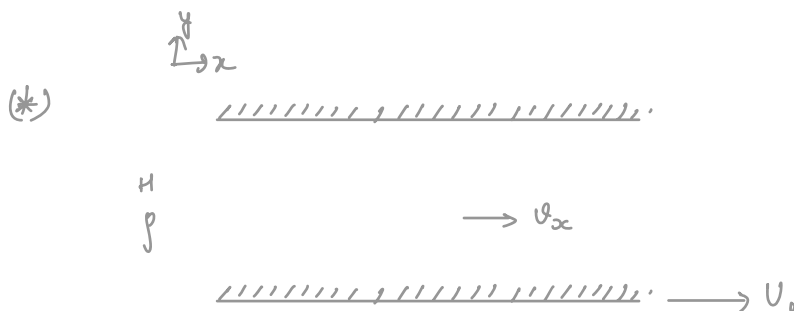
$$\tau_{xy} = -\eta \left(\frac{\partial v_x}{\partial y} \right)$$

Now writing governing equation for the above problem (2D, incompressible)

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right) + \rho g_x$$

(fully developed along x-direction) (not pressure driven) (body force only) in z-direction

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}$$



Our target is to find

$$v_x(y, t)$$

↓
velocity profile

Assumptions

Laminar flow
2D flow

fully developed flow
No slip

For newtonian fluid $\Rightarrow \tau = -\mu \frac{\partial v_x}{\partial y}$

We know

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}$$

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

ν : kinematic viscosity /
momentum diffusivity
 $= \frac{\mu}{\rho}$

Boundary conditions

$$y=0, v_x = U_0, \forall t$$

$$y=H, v_x = 0, \forall t$$

Initial condition

$$t=0, v_x = 0, \forall y$$

"Self similar"

$$\nu = \frac{\mu}{\rho} \rightarrow \text{kinematic viscosity}$$

Characteristic Value (order of magnitude analysis)

- order of magnitude of $v_x : U_0$
- order of magnitude of time : t^*
- order of magnitude of $\partial v_x : U_0$
- order of magnitude of $\partial y : H$

We have a time scale associated with momentum transport $\sim H^2/\nu$

$$\frac{U_0}{t^*} \sim \nu \left(\frac{U_0}{H^2} \right)$$

difference we are taking twice so it's only ν not ν^2

$$t^* \sim \frac{H^2}{\nu}$$

characteristic time

If time of observation $\gg t^*$, then we have steady state.

After analysis: $v_x \sim U_0, y \sim H, t \sim \frac{H^2}{\nu}$

Now we will perform non-dimensionalization

$$v_x^* = \frac{v_x}{U_0}$$

$$y^* = \frac{y}{H}$$

$$t^* = \frac{t}{(H^2/\nu)}$$

substituting in

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

$$\left(\frac{U_0}{H^2/\nu} \right) \frac{\partial v_x^*}{\partial t^*} = \frac{\nu U_0}{H^2} \frac{\partial^2 v_x^*}{\partial y^{*2}}$$

we are interested in that time domain when disturbance has reached top, so complete δ instead of ϵ

$$\Rightarrow \boxed{\frac{\partial v_x^*}{\partial t^*} = \frac{\partial^2 v_x^*}{\partial y^{*2}}}$$

$$\delta \sim \sqrt{\nu t}$$

↳ momentum diffusion length scale

(*) As long as we are close to $t^* = \frac{t}{\left(\frac{H^2}{\nu}\right)}$ then order of magnitude of LHS = 1

(*) If t is very large as compared $\frac{H^2}{\nu}$ then we drop the term $\frac{\partial v_x^*}{\partial t^*}$ as $\partial t^* \gg \partial v_x^*$ thus giving us $\frac{\partial^2 v_x^*}{\partial y^{*2}} = 0$ (STEADY STATE)

↓
logical !!

Boundary conditions $v_{x^*} = v_x / U_0 = U_0 / U_0 = 1$

Need these for analytical solution

Boundary Conditions: $\left\{ \begin{array}{ll} y^* = 0 & v_{x^*} = 1 \quad \forall t^* \\ y^* = 1 & v_{x^*} = 0 \quad \forall t^* \end{array} \right\}$

Initial condition: $\left\{ t^* = 0, v_{x^*} = 0, \forall y^* \right\}$

Initial condition

↳ $\forall y \rightarrow$ except boundaries

The following equation (proposed solution):- $\uparrow$ Proposed solution (a claim)

$$v_x^*(y, t) = v_{ss}(y) + v_t(y, t)$$

↓ depends on both y, t

steady state

↓ independent of y

unsteady state (time dependent part)

depends on both y, t

→ All variables are non-dimensional (for convenience we drop the $*$ symbol).

(*) $\frac{\partial^2 v_{ss}}{\partial y^2} = 0$

$v_{ss}(y) = ay + b$

Steady state part.

Applying boundary condition

$$y = 0, v_{ss} = 1$$

$$y = 1, v_{ss} = 0$$

$$v_{ss}(y) = 1 - y$$

for different t we get same curve for

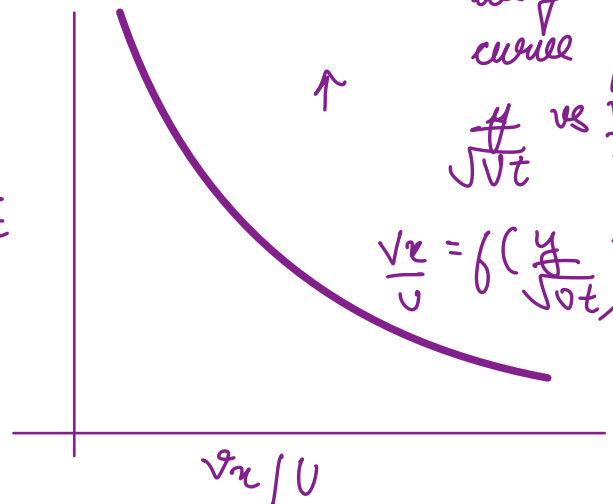
$$\frac{y}{\sqrt{\nu t}} \text{ vs } \frac{v_x}{U}$$

$$\frac{v_x}{U} = f\left(\frac{y}{\sqrt{\nu t}}\right)$$

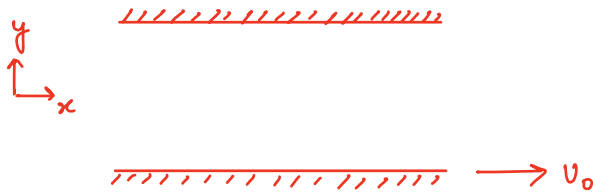
$v_x = f(\eta)$

$\eta = y / \sqrt{\nu t}$

→ space & time combined in a single variable



PRITHVI RAJ SURYAVANSHI (20CH3FP52)



$$\frac{\partial v_x}{\partial t} = \frac{\partial^2 v_x}{\partial y^2}$$

At steady state
 $v_x^* = v_{ss}$
 & boundary conditions
 will also remain
 same.

$$v_x = v_{ss} + v_t$$

Our aim is to satisfy

- ① Governing eq'n
- ② Boundary eq'n
- ③ Initial condition

SS

$$\frac{\partial^2 v_{ss}}{\partial y^2} = 0 \Rightarrow v_{ss} = ay + b$$

$$v_{ss} = 1 - y \quad (\text{The B.C. for } v_{ss} \text{ \& Total velocity } (v_x) \text{ will remain the same})$$

$$\therefore v_x = v_{ss} + v_t$$

$$v_x(y, t) = 1 - y + v_t(y, t)$$

Now substituting in governing eq'n

$$\frac{\partial v_x}{\partial y} = 0 - 1 + \frac{\partial v_t}{\partial y}$$

$$\frac{\partial^2 v_x}{\partial y^2} = \frac{\partial^2 v_t}{\partial y^2}$$

We know from NS

$$\frac{\partial v_x}{\partial t} = \frac{\partial^2 v_x}{\partial y^2}$$

$$\frac{\partial v_t}{\partial t} = \frac{\partial^2 v_t}{\partial y^2}$$

Imp $\left(\frac{\partial v_t}{\partial t} = \frac{\partial^2 v_t}{\partial y^2} \right) \rightarrow v_t = v_{x,t}(y, t)$

B.C

① $y = 0 \Rightarrow v_x = 1$ → non-dimensional term (all variables are non-dimensional)
 $\Rightarrow 1 - y + v_t = 1$
 $\Rightarrow 1 - 0 + v_t = 1 \Rightarrow v_t = 0, \quad \forall t$ $t \sim \frac{H^2}{\nu}$

② $y = 1 \Rightarrow v_x = 0$
 $\Rightarrow 1 - y + v_t = 0$

P.D with homogeneous boundary condition

$v_t = f(y)g(t) \rightarrow$ valid for homogeneous boundary condition when '0' at both boundary

$$\Rightarrow v_t = 0, \quad \forall t$$

I.C

$$\textcircled{1} \quad t=0, \quad v_x = 0 \quad \forall y$$

$$1-y+v_t = 0$$

$$v_t = y-1$$

not physically realizable

Now applying separation of variable: $v_t = f(y)g(t)$

$\hookrightarrow$ substituting in eq'n (*)

$$f \frac{\partial g}{\partial t} = g \frac{\partial^2 f}{\partial y^2}$$

$$f g' = g f''$$

$$\frac{g'}{g} = \frac{f''}{f} = c$$

only func'n
of t

only func'n
of y

$\Rightarrow$ Only possible when both
are constant

$$\frac{1}{g} \frac{dg}{dt} = c$$

$$\Rightarrow \frac{dg}{g} = c dt \Rightarrow \ln g = ct + d$$

$$g = e^{ct+d}$$

$$g = Ae^{ct}$$

imp $\checkmark$ c = always a negative number
so that after a long time
 $g \rightarrow 0$ (converges)

Therefore, writing it in a new form

$$\frac{1}{g} \frac{dg}{dt} = -c^2$$

$$g = Ae^{-c^2 t}$$

$-c^2 \rightarrow$ so that always negative

$$\text{Now,} \quad \frac{g'}{g} = \frac{f''}{f} = -c^2$$

$$\frac{1}{f} \frac{d^2 f}{dy^2} = -c^2 \Rightarrow \frac{d^2 f}{dy^2} + c^2 f = 0$$

$$\Rightarrow f = B \sin(cy) + D \cos(cy)$$

$$(*) \quad y=0, \quad \forall t=0 \quad \forall t$$

always only possible when $f(y)=0$

$$f(y)g(t) = 0 \quad \forall t$$

$$\Rightarrow f(y) = 0$$

, $g(t)$ can be zero for some t , but not all t .

$$y=1, \quad f(y) = 0 \quad \forall t$$

$$\text{At } y=0, y=1, \quad f=0$$

$$\text{At } y=0 \Rightarrow 0 = 0 + D \Rightarrow D=0$$

$$\text{At } y=1 \Rightarrow 0 = B \sin c \Rightarrow \sin c = 0 \Rightarrow c = n\pi$$

$$f'' = -c^2 \Rightarrow f(y) = B_m (\sin c_n y)$$

$$c_n = n\pi$$

$$f(y) = \sum_{-\infty}^{\infty} B_m \sin(c_n y)$$

$$v_t(y, t) = \sum A_n e^{-c_n^2 t} B_n \sin(c_n y)$$

$$= \sum D_n e^{-c_n^2 t} \sin(c_n y)$$

general solution

also a solution

linear combination

$$v_x = v_{ss} + v_t$$

$$= 1-y + \sum_{n=-\infty}^{\infty} D_n e^{-c_n^2 t} \sin(c_n y)$$

##) 02/02/2023

////////////////////

//////////////////// $\rightarrow v^*$

$$v_{sc}(y^*, t^*) = \left(\sum_{n=-\infty}^{\infty} D_n \exp(-n^2 \pi^2 t^*) \sin(n\pi y^*) \right) + (1-y)$$

As of now we have applied B.C's but have not applied I.C.

So now applying Initial Conditions:-

$$t=0, v=0$$

$$\bar{y}^{-1} = \sum_{n=-\infty}^{\infty} D_n \sin(n\pi y^*)$$

coupling negative & positive counterparts

$$\bar{y}^{-1} = \sum_{n=1}^{\infty} D_n \sin(n\pi y^*)$$

y^* is written as y after this step

$$\sin(-\theta) = -\sin \theta$$

$$C_m = m\pi$$

If $+m_1$ is a soln then D_{m_1} be the coefficient, so for $-m_1$, coefficient will be $-D_{-m_1}$

$$\sum_{n=0}^{\infty} (D_{m_1} - D_{-m_1}) \sin(C_{m_1} y) e^{-C_{m_1}^2 t}$$

$$(\bar{y}-1) \sin(m\pi y) = \sum_{n=1}^{\infty} D_n \sin(n\pi y) \sin(m\pi y)$$

$$\int_0^1 (\bar{y}-1) \sin(m\pi y) dy = \sum_{n=1}^{\infty} \int_0^1 \sin(n\pi y) \sin(m\pi y) dy$$

$\nearrow 0 (m \neq n)$
only surviving term $\Rightarrow n=m$ only $m=n$ will survive

$$\int_0^1 (\bar{y}-1) \sin(n\pi y) dy = D_n \int_0^1 \sin^2(n\pi y) dy$$

$$D_n = \frac{-2}{n\pi} \quad (\text{after solving})$$

$$u_x^*(y^*, t^*) = \left[\sum_{n=1}^{\infty} \left(\frac{-2}{n\pi} \right) \exp(-n^2 \pi^2 t^*) \sin(n\pi y^*) \right] + (1-\bar{y})$$

$t^* = t/(H^2/v)$

(#) A function of the form $P_n = \sin(C_n y)$ will satisfy $P_m = \sin(C_m y)$

If $m \neq n$

$$P_m \frac{d^2 P_n}{dy^2} + C_n^2 P_n P_m = 0$$

$$\int_0^1 P_m P_n dy = 0$$

Integrating

$$\int_0^1 P_m \frac{d^2 P_n}{dy^2} dy + c_n^2 \int_0^1 P_n P_m dy = 0$$

$$P_m \frac{dP_n}{dy} \Big|_0^1 - \int_0^1 \frac{dP_m}{dy} \frac{dP_n}{dy} dy + c_n^2 \int_0^1 P_n P_m dy = 0$$

using same procedure for P_m

$$\begin{aligned} & - \int_0^1 \frac{dP_m}{dy} \frac{dP_n}{dy} dy + c_n^2 \int_0^1 P_n P_m dy = 0 \\ - & - \int_0^1 \frac{dP_n}{dy} \frac{dP_m}{dy} dy + c_m^2 \int_0^1 P_m P_n dy = 0 \\ \hline & (c_n^2 - c_m^2) \int_0^1 P_m P_n dy = 0 \end{aligned}$$

Dimensional form of the equation

$$\frac{v_x(y, t)}{v_0} = \sum_{n=1}^{\infty} \left(\frac{-2}{n\pi} \right) \exp\left(-n^2 \pi^2 \frac{t}{H^2/\nu} \right) \sin\left(n\pi \frac{y}{H} \right) + \left(1 - \frac{y}{H} \right)$$

NOTE ** = velocity profile (spatial variance, for a given t)

#) In case we have condition in terms of T_0 instead of v_0 .

||||| $\rightarrow T_0$

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}$$

||||| $\rightarrow T_0$

we will try and convert the eqn in terms of T , using the CONSTITUTIVE RELATION

$\Downarrow$

$$T = -\mu \frac{\partial v_x}{\partial y}$$

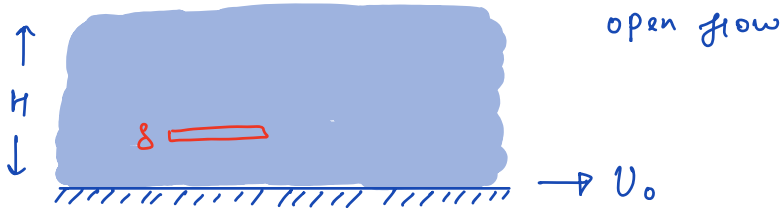
multiplying with $-\mu$

$$-\mu \frac{\partial v_x}{\partial t} = \frac{\mu}{f} \frac{\partial^2}{\partial y^2} (-\mu v_x)$$

$$\frac{\partial}{\partial y} \left(-\mu \frac{\partial v_x}{\partial t} \right) = \frac{\partial}{\partial y} \left(\frac{\mu}{f} \frac{\partial}{\partial y} \left(-\mu \frac{\partial v_x}{\partial y} \right) \right)$$

$$\frac{\partial \tau}{\partial t} = \frac{\mu}{f} \frac{\partial^2 \tau}{\partial y^2}$$

(#)



If $\delta \ll H$, semi-infinite domain.

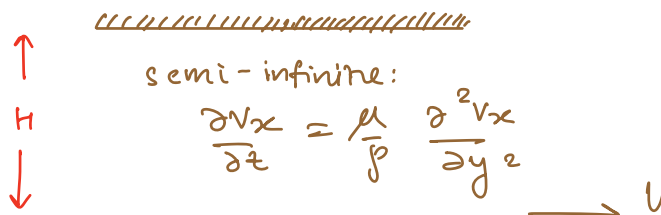
δ : region upto which fluid has moved, or momentum has been transferred.

09/02/2023

Stoke's first problem: flat plate being pulled with constant velocity



(unidirectional $v_x \neq 0, v_y = 0, v_z = 0$)



$$\overline{\overline{y=0, v_x = U, \forall t > 0}}$$

(Ans.)

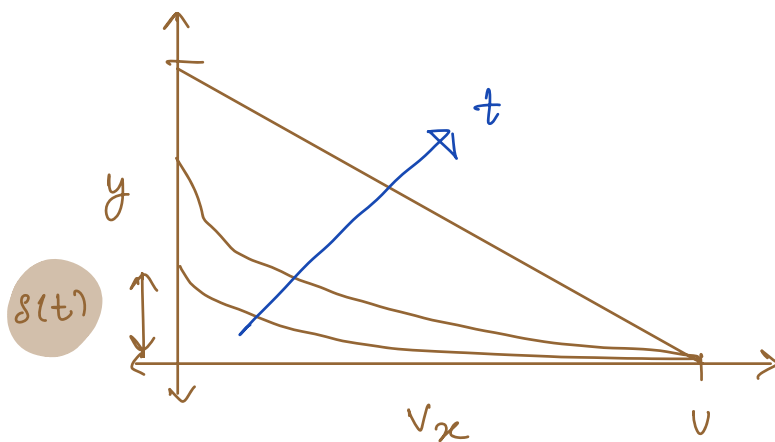
$t \sim \frac{H^2}{\nu} \rightarrow$ only if this is satisfied we can consider H

If $t \ll \frac{H^2}{\nu} \rightarrow y \rightarrow \infty, v_x = 0, \forall t$

early times

upper boundary condition
 $\rightarrow H$ will be very large as compared to the disturbance.

velocity profile:



$$t_{obs} \ll \frac{H^2}{\nu}$$

$s(t)$: characteristic length scale for early times

$$f(\eta) = \frac{v_x}{U_0} \quad ; \quad \eta = \frac{y}{s(t)}$$

Note: we are assuming fully developed in x -direction.

goal is to convert all variables to f & η .

$$\underline{\partial v_x} = \left(\underline{\partial v_x} \right) / \left(\underline{\partial \eta} \right)$$

$$\frac{\partial}{\partial t} \left(\frac{\partial \eta}{\partial t} \right) \left(\frac{\partial \eta}{\partial t} \right)$$

$$\frac{\partial v_x}{\partial t} = \left(v_0 \frac{\partial f}{\partial \eta} \right) \left(-\frac{\gamma}{s^2} \delta' \right)$$

$$\frac{\partial \eta}{\partial t} = (\gamma) \left(-\frac{1}{s^2} \right) \frac{d\delta}{dt}$$

$$\frac{\partial v_x}{\partial \eta} = v_0 \frac{\partial f}{\partial \eta}$$

$$\frac{\partial v_x}{\partial y} = \frac{\partial v_x}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = v_0 \frac{\partial f}{\partial \eta} \left(\frac{1}{s} \right)$$

$$\frac{\partial^2 v_x}{\partial y^2} = \frac{\partial}{\partial \eta} \left(\frac{\partial v_x}{\partial y} \right) \cdot \frac{\partial \eta}{\partial y}$$

$$= \frac{\partial}{\partial \eta} \left(\frac{v_0}{s} \frac{\partial f}{\partial \eta} \right) \cdot \frac{\partial \eta}{\partial y}$$

$$= \frac{v_0}{s} \frac{\partial^2 f}{\partial \eta^2} \downarrow \frac{1}{s}$$

$$= \frac{v_0}{s^2} \frac{\partial^2 f}{\partial \eta^2}$$

ISKE BAAD PROOF DEKH LENA !!!!